

Machine learning center-specific models show improved IVF live birth predictions over US national registry-based model

Corresponding Author: Dr Mylene Yao

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

In this paper, the authors present a study on the difference between using clinic-specific models and a general model for IVF-LB predictions. This topic is very relevant and there are few studies that have studied the difference between in-house models and general models. The paper is well-structured and well-written.

The abstract is difficult to read when the MLCS-1/2 models are not explained. I suggest adding a brief explanation.

Introduction

Reference to LinkedIn posts and personal communication should be avoided. The final section should be reformulated. The focus should be on the study hypothesis and the clinical endpoints. Metrics should not be the “primary outcome” as it is only the method used to demonstrate the hypothesis. Please also remove the sentence with AMH as this is too detailed for the introduction. In addition, the last 4 lines should be removed and/or moved to the discussion. Also, clarify the hypothesis and end-points for the drift sub-study.

Methods

Please verify that the data is actually anonymized. When the data sets contain that many model predictors it seems unlikely that it will not be possible to identify a patient in a GDPR sense. Most likely the data is only pseudo-anonymized.

Why is the term “de novo model validation test set” used? Suggest to follow the TRIPOD+AI terminology and use internal validation data.

There should be a mention of the framework for doing the statistics.

Results

There is missing a descriptive table with differences between clinics. This must (at least) include the parameters that are relevant for the SART predictions (AMH, Age etc.).

The reported AUC improvements compared to age-only control seem very high. An 41% improvement over an AUC = 0.5 is 0.70. Does that imply that the age-only control has an AUC of less than 0.5? For tables 2 and 3 I would strongly recommend that this is reported per clinic with the use of 95% C.I. and a paired de-long test for AUC. Note that references to table 2 and 3 in several places are incorrect.

Regarding model drift the authors state that this was not detectable. Would suggest that a non-paired AUC test is performed per clinic to validate this statement.

For table 3 it is not possible to see the results for the 40%, 60% and 70% threshold, so would remove the mentioning in the text or add it to the table. When the statistical difference for PR AUC is mentioned (legend should include the test method) the AUC-ROC should also include a statistical test.

The authors might also consider to only use DNMV1 or DNMV2 internal validation data as there are limited value with presenting results from both DNMV1 and 2. From all practical points, the results are identical and only add complexity to understanding the results. Otherwise, it is suggested that all results for one of the sets are moved to supplementary data.

For figure 1B I would suggest making a new figure with real calibration plots for both models using 4-5 bins. This should be done for both individual clinics and overall. The calibration plots should include the data distribution (TRIPOD example here (figure 8): <https://www.acpjournals.org/doi/10.7326/M14-0698>). Showing calibration plots is also necessary to claim that the study adheres to TRIPOD-AI.

Discussion

When discussing the reclassification it would be useful to add an analysis of if the improved classification is general for all 6 clinics or mainly observed for a few clinics.

PGT-A is only briefly mentioned and is not excluded. However, assume that the amount of PGT-A testing would have a large impact on the performance of the SART.

At several places in the discussion, the text gets too commercial. For example, the sections "Like many innovative products...", "MLCS-based counseling reports..." is not relevant for the study. The commercial side of the discussion should be toned-down.

The discussion should also include a section on the limitations of the study.

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)

In their manuscript, the authors demonstrate that using machine learning in 6 US fertility centers, a better predictive model is developed when compared to the SART IVF live birth prediction model. As a result, this might lead to better informing fertility patients' treatment decisions. The paper is well written and although I am not an expert on machine learning, the authors managed to guide me through the manuscript resulting in understanding the design and results from their study.

The manuscript however requires revision in order to address some concerns.

Firstly, the author refer to 'true value based IVF care at scale, through transparent IVF pricing and live birth outcomes specific to each fertility center'. However, they fail to demonstrate this, since only aggregated data are presented without presenting data, including financial, of the individual centers. Either the authors should provide data concerning the 6 individual centers, including the number of patients per center included in the model, or the topic of the manuscript should be the comparison of their model with the SART model. When referring to value based IVF care, reference should be made to the publications by Porter and Teisberg (e.g. Redefining Health Care, 2006, Harvard Business School Press). And if personalized counseling based on individual baseline characteristics ('patient like me') is the main purpose, I suggest the authors take note of the publication by Bensink et al (BMJ Open, 2023).

Secondly, the authors need to describe feature selection in more detail since this can have great influence on the outcome of the model. Questions that at least should be addressed are for instance: which features were selected, was the same feature set used in all centers or individualized per center, what was the relative contribution of each feature in the model. Figure 1 is not sufficiently self-explanatory. Can the authors also provide data on reclassification per center in order demonstrate consistency of the MLCS2 model across centers? What are the limitations of their findings?

And finally, can the authors discuss the clinical relevance of the statistically significant, although relatively small increase in predicting live births using machine learning? Can the authors explore how to further improve the model? And with regard to value based healthcare, how can the authors' findings assist in determining in which center the best value for money for any individual patient s delivered?

(Remarks on code availability)

The code was not available for review.

Reviewer #3

(Remarks to the Author)

The study "Predicting IVF live birth probabilities using machine learning, center-specific models: validation results and potential benefits over national registry-based models" presents a well-documented comparative analysis of machine learning-based center-specific (MLCS) models and the SART national model for IVF live birth prediction, but some technical concerns hinder its conclusions.

First, the authors rely on gradient-boosted machines (GBMs) but do not provide sufficient detail on important aspects such as feature selection, hyperparameter tuning, and model interpretability. Without comparisons to other machine learning for structured data, such as Random Forests, CatBoost, or lightGBM, it is unclear whether the performance gains arise from the MLCS framework, gradient boosting itself, or merely the inclusion of additional predictors.

The focus on Precision-Recall AUC and F1 Score is appropriate for imbalanced data, but the authors do not adequately justify the exclusion of metrics like calibration curves and Brier scores. A stronger evaluation would also involve analyzing calibration across centers.

The involvement of Univfy Inc. introduces a potential conflict of interest, as the MLCS framework is a proprietary product. The refusal to share data or details regarding its preprocessing pipelines raises concerns about reproducibility and independent

verification of the results. Additionally, the framing of SART as an “outdated” baseline without offering clear well-defined alternatives, such as well-defined ML models or recalibrated SART versions, further skews the analysis.

(Remarks on code availability)

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Dear Authors,

Thank you for your thorough revision of your manuscript in response to my review. I am very pleased to see that you have addressed each of my comments with care and precision. This among others includes the updates with including per clinic results, the inclusion of calibration plots which are essential for demonstrating adherence to TRIPOD-AI guidelines and the discussion of reclassification and acknowledgement of study limitations.

The manuscript is now significantly improved in terms of clarity and adherence to reporting guidelines.

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)

I wish to compliment the authors for taking note of the concerns that I have raised, thereby substantially improving the manuscript. I can therefore recommend the manuscript acceptable for publication, thereby recognizing that the ultimate decision is up to the editors discretion.

Just one small remark which I would like the authors to take note of, perhaps in future work. In reproductive medicine, the live birth rate is no longer the sole or most important outcome measure. Outcomes from all of the tiers as described by Porter matter to patients.

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)

The revisions made to the manuscript were satisfactory, and it is now ready for publication.

(Remarks on code availability)

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REVIEWER COMMENTS AND POINT-BY-POINT RESPONSES (IN BLUE)

Dear Reviewers,

Please note that as mentioned in the cover letter, we revised the title to specify that this study compared MLCS models against the US national registry-based model only. In our original submission, the title said “national registry-based models” which was a generalization that was unintended. Since we only compared against one national registry model, we should limit the title to the exact comparison.

In addition to the changes mentioned in response to reviewers’ comments, we also added information to comply with TRIPOD and Nature journals’ policies re:

- model fairness, non-exclusion of patients based on race, ethnicity, religion or financial/insurance coverage status;
- explicitly stated the users of each type of model (whether they were providers or patients or both);
- the biological sex of the patients was female due to the nature of the IVF treatment with relevant information from male partner included in the data analysis and modeling
- no patients were excluded due to not having a male partner (in the event that donor sperm was used)

Thank you!

Sincerely,

Mylene Yao

Reviewer #1 (Remarks to the Author):

In this paper, the authors present a study on the difference between using clinic-specific models and a general model for IVF-LB predictions. This topic is very relevant and there are few studies that have studied the difference between in-house models and general models. The paper is well-structured and well-written.

The abstract is difficult to read when the MLCS-1/2 models are not explained. I suggest adding a brief explanation.

A brief explanation of MLCS1 and MLCS2 models have been added to the abstract.

Introduction

Reference to LinkedIn posts and personal communication should be avoided. The final section should be reformulated. The focus should be on the study hypothesis and the clinical endpoints. Metrics should not be the “primary outcome” as it is only the method used to demonstrate the hypothesis. Please also remove the sentence with AMH as this is too detailed for the introduction. In addition, the last 4 lines should be removed and/or moved to the discussion. Also, clarify the hypothesis and end-points for the drift sub-study.

The LinkedIn and personal communication references have been removed from the Introduction.

The final section of the Introduction has been revised to focus on study hypothesis and clarified that model metrics were used to evaluate the models rather than serving as "primary outcome".

The sentence regarding AMH has been removed from the Introduction.

We have moved the last 4 lines of Introduction to Discussion.

We have clarified the hypothesis and end-points (ROC-AUC, posterior log odds ratio compared to age control model) for testing external model validation using the external, out-of-time test set for the MLCS1 models. We agree the term data drift was imprecise.

We have now broken down that part of the investigation to test external model validation (as an example of concept drift) separately from comparison of statistical distributions of clinical variables (as an example of data drift).

- Our main focus was to determine whether previously validated models continued to be appropriate for clinical utility in the time period contemporaneous with model usage. We have clarified these points in the last paragraph of the Introduction, 4th-5th paragraphs of Results, and the 10th paragraph of Discussion. Methods are described in the Methods section.
- Regarding data drift, we tested the statistical distribution of key clinical variables by two-sample Kolmogorov-Smirnov test (in the case of continuous variables) and Chi-square test (in the case of frequency of diagnoses and sperm-related variables). See Supplementary Tables 3A-F for these statistical distributions and statistical test results at the clinic level.

Methods

Please verify that the data is actually anonymized. When the data sets contain that many model predictors it seems unlikely that it will not be possible to identify a patient in a GDPR sense. Most likely the data is only pseudo-anonymized.

The methods section has been updated to state that data was de-identified, rather than anonymized.

Why is the term “de novo model validation test set” used? Suggest to follow the TRIPOD+AI terminology and use internal validation data.

We have now applied TRIPOD+AI internal and external validation terminology. Since "external validation" may be interpreted to mean test data from a different geographical location (rather than from a different time period of the same clinic), we explicitly included "in-time" and "out-of-time" to further specify the nature of the test data (see Table 1).

We used the term “de novo model validation test set” (DNMV) because the DNMV test set is not 100% identical to the 6 clinic-specific test sets that were originally used to perform internal cross-validation during the model creation process. The DNMV test sets were adapted to meet the input criteria of the SART model and the MLCS2 models. We agree the name was a bit clunky and our intention was just to convey to readers that the test set was derived from but not the same as the model validation test sets, hence the word "de novo".

We have incorporated TRIPOD+AI terminology of internal validation in the “Creating center-specific, de novo model validation test sets (DNMV1 and DNMV2) to enable comparison of MLCS2 and SART models” section.

There should be a mention of the framework for doing the statistics.

We have now provided more details regarding statistical tests in the Results, Methods and Table descriptions.

Results

There is missing a descriptive table with differences between clinics. This must (at least) include the parameters that are relevant for the SART predictions (AMH, Age etc.).

We have added a descriptive table showing the summary statistics (mean and standard deviation) for key numeric variables (e.g. female age, BMI, AMH, etc.) and frequency for key diagnoses and sperm-related variables for each of the 6 center-specific test sets making up the DNMV1 test sets, which were used to evaluate and compare SART and MLCS2 models. See Table 3.

We have added a similar descriptive table for MLCS1 internal and external (live model validation) test sets for each of the 6 centers. See Supplementary Table 3A-F.

The reported AUC improvements compared to age-only control seem very high. An 41% improvement over an AUC = 0.5 is 0.70. Does that imply that the age-only control has an AUC of less than 0.5? For tables 2 and 3 I would strongly recommend that this is reported per clinic with the use of 95% C.I. and a paired de-long test for AUC. Note that references to table 2 and 3 in several places are incorrect.

The age model AUC results from internal cross validation were indeed lower than those of the MLCS models. The age model AUC median and IQR have been added to Table 2. We revised Table 2 to show more explicitly each comparison and the median AUCs and IQRs used for each comparison and where appropriate, we also indicated the statistical significance.

Per-clinic versions of Table 2 have been created and added to the supplemental data (Supplementary Tables 1 & 2). For the comparison of each center's MLCS2 vs Age model, unweighted AUC with 95% CI values and results for paired DeLong tests are provided at the center level. (Supplementary Table 1.)

Please note that in Supplementary Tables 1, we show the weighted AUC values for MLCS1 and MLCS2 models and their corresponding Age models next to the unweighted AUC values. We typically use weighted AUC values in cross validation tests to avoid over-representation of clinical profiles of a small group of patients with recurrent IVF cycles. However, calculating the 95% confidence interval and the application of DeLong's Test necessitated using unweighted AUCs (based on the R packages that we found). Therefore, we showed both weighted and unweighted AUCs in the supplementary table.

In Supplementary Table 2 which concerns comparison of MLCS1 internal vs. external validation, we showed only the unweighted values for the specific purpose of performing the DeLong's Test and the p-values at the center level. We did not show the weighted values for MLCS1 CV results for each center in this table since they are available in Supplementary Table 1 and because the comparison between the external validation LMV data set ROC-AUC and the internal CV weighted ROC-AUC values across centers was not statistically significant (Table 2). Comprised of only first cycles, the external validation LMV data set had no difference between its weighted and unweighted ROC-AUC.

References for all tables have been reviewed and updated where appropriate.

Regarding model drift the authors state that this was not detectable. Would suggest that a non-paired AUC test is performed per clinic to validate this statement.

As stated above, Supplementary Table 2 now shows unpaired AUC tests were performed per clinic to demonstrate that model performance did not decline when externally validated. Those results indicated that the model was valid for patients who received care after the model was cross-validated and implemented.

For table 3 it is not possible to see the results for the 40%, 60% and 70% threshold, so would remove the mentioning in the text or add it to the table. When the statistical difference for PR AUC is mentioned (legend should include the test method) the AUC-ROC should also include a statistical test.

F1 scores at the 40%, 60%, and 70% thresholds have been added to the previous Table 3, now Table 4.

The statistical test has been added to the table caption for the previous Table 3, now Table 4. We compared F1 scores at the 50% threshold only. We did not perform statistical comparison at the other thresholds to avoid multiple testing error.

The authors might also consider to only use DNMV1 or DNMV2 internal validation data as there are limited value with presenting results from both DNMV1 and 2. From all practical points, the results are identical and only add complexity to understanding the results. Otherwise, it is suggested that all results for one of the sets are moved to supplementary data.

We agree and have removed the DNMV2 results because they do not add to the manuscript.

For figure 1B I would suggest making a new figure with real calibration plots for both models using 4-5 bins. This should be done for both individual clinics and overall. The calibration plots should include the data distribution (TRIPOD example here (figure 8): <https://www.acpjournals.org/doi/10.7326/M14-0698>). Showing calibration plots is also necessary to claim that the study adheres to TRIPOD-AI.

Calibration plots, including data distribution (following TRIPOD-AI Figure 8 example), for models aggregated across clinics were added to Figure 1 as Figure 1A. Calibration plots, including data distribution, were also created at the clinic level and shown in Supplemental Figure 2.

Discussion

When discussing the reclassification it would be useful to add an analysis of if the improved classification is general for all 6 clinics or mainly observed for a few clinics.

A reclassification table was created for the aggregate test data (from 6 centers) (Table 5) and for each clinic (Supplementary Table 5A-F) in order to comment on the generality of the improved classification within the context of these 6 centers. The pattern of MLCS2 placing more patients in higher prognostic categories compared to SART model was observed in each center's reclassification table.

In the original submission, we had reported that MLCS2 models showed improved net reclassification index (NRI) over the SART model. Although that finding was favorable to MLCS2, we decided to remove the NRI metric from the paper because the interpretation of the metric is controversial in the literature and our findings do not depend on it nor are they affected by it. We noted the absence of formal statistical reclassification analyses in the Results section.

PGT-A is only briefly mentioned and is not excluded. However, assume that the amount of PGT-A testing would have a large impact on the performance of the SART.

In the Discussion section (3rd paragraph from the end of Discussion), we have now added the point that neither MLCS nor SART model specifically addressed "add-ons" such as PGT-A and ICSI, procedures that are chosen by provider-patient shared decision-making. We commented on how add-ons may be addressed outside of the models.

In the 2nd last paragraph of Discussion, we also mentioned how the use of MLCS prognostic framework may be applied to investigate the impact of add-ons in the future.

At several places in the discussion, the text gets too commercial. For example, the sections "Like many innovative products...", "MLCS-based counseling reports..." is not relevant for the study. The commercial side of the discussion should be toned-down.

Your point is well taken. Our original intention was to provide full transparency that there is a cost to providers to use MLCS model to support patient counseling, so that the costs and benefits of MLCS model usage are conveyed to the readers. We had no intention of coming across as commercial, so to err on the side of caution, we have removed that paragraph.

The discussion should also include a section on the limitations of the study.

A section on limitations has been added to the discussion. See the 3rd, 4th, and 5th paragraphs from the end of Discussion, addressing various limitations of the study, starting with, "The key limitation..."

Reviewer #2 (Remarks to the Author):

In their manuscript, the authors demonstrate that using machine learning in 6 US fertility centers, a better predictive model is developed when compared to the SART IVF live birth prediction model. As a result, this might lead to better informing fertility patients' treatment decisions. The paper is well written and although I am not an expert on machine learning, the authors managed to guide me through the manuscript resulting in understanding the design and results from their study.

The manuscript however requires revision in order to address some concerns.

Firstly, the author refer to 'true value based IVF care at scale, through transparent IVF pricing and live birth outcomes specific to each fertility center'. However, they fail to demonstrate this, since only aggregated data are presented without presenting data, including financial, of the individual centers. Either the authors should provide data concerning the 6 individual centers, including the number of patients per center included in the model, or the topic of the manuscript should be the comparison of their model with the SART model. When referring to value based IVF care, reference should be made to the publications by Porter and Teisberg (e.g. Redefining Health Care, 2006, Harvard Business School Press). And if personalized counseling based on individual baseline characteristics ('patient like me') is the main purpose, I suggest the authors take note of the publication by Bensink et al (BMJ Open, 2023).

This study's objective was to test whether machine learning center-specific (MLCS) models performed better than a national registry-based model using model metrics.

We have now made more explicit how improved model metrics such as PR-AUC and F1 scores directly impact ability of the model to support value-based IVF pricing. (See the 3rd, 5th, and 6th paragraphs in Discussion.) We have cited references (24, 55) to provide more comprehensive information including illustration of value-based pricing:

- Reference #24: Yao et al., Semin Reprod Med 2024. Patient-centric in vitro fertilization prognostic counseling using machine learning for the pragmatist. <https://www.thieme-connect.com/products/ejournals/html/10.1055/s-0044-1791536>
- Reference #55: Univfy. Sample ML-based IVF refund program cost comparison. Accessed Feb. 8, 2025 at: <https://www.univfy.com/research/sample-ivf-refund-program-cost-comparison>

A fuller report/discussion of the value-based pricing implementations and their real-world results is beyond the scope of this article because those financial results are not easily adapted to meet the requirements of IRB-exempt research protocols. Also, it would not be possible to present financial implementations of all 6 centers at the same time due to center-specific requirements. We do aim to report individual case studies after meeting IRB requirements.

We have also cited the reference (Porter and Teisberg, 2006) and also added more context to describe the unique provider and patient needs from value-based IVF pricing.

Separately, we have noted Bensink et al BMJ Open 2023 – thank you!

We have not cited Bensink et al, because the manuscript is already quite lengthy. Bensink et al uses a database-driven approach rather than prediction modeling. Clinical variables and relative weightings (if any) were not determined based on objective testing. Consequently, there is an increased risk of overfitting if some variables were not predictive of treatment outcome or the appropriate relative weighting was not applied.

Secondly, the authors need to describe feature selection in more detail since this can have great influence on the outcome of the model. Questions that at least should be addressed are for instance: which features were selected, was the same feature set used in all centers or individualized per center, what was the relative contribution of each feature in the model. Figure 1 is not sufficiently self-explanatory. Can the authors also provide data on reclassification per center in order demonstrate consistency of the MLCS2 model across centers? What are the limitations of their findings?

Feature Selection

Feature selection details are highlighted in Supplemental Figure 1, which is referenced in the “Model predictors” sub-section of the “Methods” section. This figure lists features that were selected at each center, whether the common features were selected across centers, and the relative contribution of each feature in the MLCS2 model at each center. The “Model predictors” sub-section of the Methods section (“Feature sets, model predictors and relative importance of model predictors”) has also been revised for greater clarity.

Figure 1: More details have been added to Figure 1’s caption.

Reclassification:

A reclassification table was created for the aggregate test data (from 6 centers) (Table 5). We have now added a reclassification table for each clinic (Supplementary Tables 5A-F) in order to comment on the generality of the improved classification.

The pattern of MLCS2 placing more patients in higher prognostic categories compared to SART model was observed in each center's reclassification table.

We further clarified that the reclassification table is provided for the sole purpose of discussing clinical context and to help translate our findings to clinicians.

In the original submission, we had reported that MLCS2 models showed improved net reclassification index (NRI) over the SART model. Although that finding was favorable to MLCS2, we decided to remove the NRI metric from the paper because the interpretation of the metric is controversial in the literature and our findings do not depend on it nor are they affected by it. We noted the absence of formal statistical reclassification analyses in the Results section.

Limitations: A section on limitations has been added to the discussion. See the 3rd, 4th, and 5th paragraphs from the end of Discussion, addressing various limitations of the study, starting with, "The key limitation..."

And finally, can the authors discuss the clinical relevance of the statistically significant, although relatively small increase in predicting live births using machine learning? Can the authors explore how to further improve the model? And with regard to value based healthcare, how can the authors' findings assist in determining in which center the best value for money for any individual patient's delivered?

The model improvements, especially the F1 score and PR AUC (Table 4) are non-trivial for a prognostic model and the improved F1 score is consistently observed at live birth prediction thresholds >40% to >70%. We created the reclassification tables (Table 5, Supplemental Tables 5A-F) and highlighted the following bullets in Results to illustrate the clinical impact:

- ~30% (1386 of 4645) of the patients had discordant LBPs (Table 5, peach or blue cells) -- meaning, MLCS2 and SART models placed 1386 patients into different prognostic categories.
- MLCS2 assigned 26% (1230 of 4645) of patients to a higher LBP category compared to the SART model (Table 5, blue cells).
- MLCS2 appropriately identified 23% and 11% of patients with $LBP \geq 50\%$ and $LBP \geq 75\%$, respectively, compared to lower LBPs given by SART.
- MLCS2 models identified ~11% (494 of 4645) of patients to have $LBP \geq 75\%$, whereas SART model did not place any patient in that group even though the actual live birth rate of the MLCS- $LBP \geq 75\%$ group was 81%.

How to further improve model:

In the last paragraph of the discussion, we suggest to “advance reproductive medicine beyond dichotomies of multicenter versus center-specific or ML versus non-ML prediction models....We hypothesize the optimal prognostic solution is the multicenter scaling of a machine learning, localized approach. One possible next-step is to study IVF LBP model metrics on a wider range of fertility centers -- larger centers, academic centers, centers in IVF coverage-mandated states and publicly funded IVF – to test whether multicenter scaling of MLCS models may bring additional benefits. We recommend expanding collaborative research efforts between public and private sectors to tackle crucial questions related to racial disparities in IVF, molecular mechanisms underlying clinical infertility and IVF failure, in order to both expand access to IVF care and improve live birth outcomes and the patients' experience.”

RE: "best value for money" -- We have elaborated on value-based IVF pricing in the 5th and 6th paragraphs in the Discussion section.

It is beyond the scope of this article to provide examples to fully illustrate those points. However, we have cited and referred to a review paper and a research supplementary webpage that address "best value for money".

- Reference #55: Univfy. Sample ML-based IVF refund program cost comparison. Accessed Feb. 8, 2025 at: <https://www.univfy.com/research/sample-ivf-refund-program-cost-comparison>

- Reference #24: Yao et al., Semin Reprod Med 2024. Patient-centric in vitro fertilization prognostic counseling using machine learning for the pragmatist. <https://www.thieme-connect.com/products/ejournals/html/10.1055/s-0044-1791536>

Reviewer #2 (Remarks on code availability):

The code was not available for review.

Code that was not available via a publicly accessible package/library and a dummy data set were uploaded to the Code Ocean profile for this paper. We will ask for help from the Editorial team to figure out how to send the Code Ocean link to reviewers.

Reviewer #3 (Remarks to the Author):

The study "Predicting IVF live birth probabilities using machine learning, center-specific models: validation results and potential benefits over national registry-based models" presents a well-documented comparative analysis of machine learning-based center-specific (MLCS) models and the SART national model for IVF live birth prediction, but some technical concerns hinder its conclusions.

First, the authors rely on gradient-boosted machines (GBMs) but do not provide sufficient detail on important aspects such as feature selection, hyperparameter tuning, and model interpretability. Without comparisons to other machine learning for structured data, such as Random Forests, CatBoost, or lightGBM, it is unclear whether the performance gains arise from the MLCS framework, gradient boosting itself, or merely the inclusion of additional predictors.

Feature selection details are highlighted in Supplemental Figure 1, which is referenced in the "Model predictors" sub-section of the "Methods" section ("Feature sets, model predictors and relative importance of model predictors"). This figure includes which features were selected at each center, whether the common features were selected across centers, and the relative contribution of each feature in the MLCS2 model at each center. The "Model predictors" sub-section has also been revised for greater clarity.

We provided the following relevant discussion:

- 8th paragraph of Discussion: We offered potential factors contributing to differential model performance between MLCS and SART models. Pragmatically, the improved performance is likely due to a combination of factors.
- 9th paragraph of Discussion:
 - We clarified that this study was not designed and would not be able to pinpoint/prove causes of differential model performance and that was not the goal.
 - We clarified that we are not concluding the center-specificity or machine learning to be the cause of the model improvement.
 - We provided rationale on why comparing ML techniques was not our main goal as there were many other considerations required for quality control.
- In the last paragraph of Discussion, we gave a brief outline of research efforts that we expect to yield important advances (instead of spending more time dissecting possible causes of improved MLCS model performance reported here).

The focus on Precision-Recall AUC and F1 Score is appropriate for imbalanced data, but the authors do not adequately justify the exclusion of metrics like calibration curves and Brier scores. A stronger evaluation would also involve analyzing calibration across centers.

We agree with the reviewer's comment that calibration curves and Brier scores should be included.

We have now added model evaluation by Brier score and calibration curve.

We have calculated the Brier score for MLCS2 and SART models for each of the 6 centers. The median Brier score and interquartile range are reported in Table 4; each center's Brier scores for

both models are shown in Supplementary Table 4. Interpretation of the Brier score analyses has also been added to the Results section.

Calibration plots, including live birth probability (LBP) distribution, for both models aggregated across clinics were added to Figure 1 as Figure 1A. Calibration plots, including LBP distribution, were also created at the clinic level and included as Supplemental Figure 2.

The involvement of Univfy Inc. introduces a potential conflict of interest, as the MLCS framework is a proprietary product. The refusal to share data or details regarding its preprocessing pipelines raises concerns about reproducibility and independent verification of the results. Additionally, the framing of SART as an “outdated” baseline without offering clear well-defined alternatives, such as well-defined ML models or recalibrated SART versions, further skews the analysis.

The datedness only related to the years of data used for training the SART model (2014-2016) with no recent updates or recalibration. We have removed the word "outdated".

The employees of Univfy Inc. provided full disclosure of potential conflict of interest.

We agree with the importance of data-sharing and also wish it were standard to share data in the IVF field. Unfortunately, sharing of clinical data is not the norm in clinical IVF research. The authors of this study and Univfy are not authorities on the lack of data-sharing in IVF but we wish to respect the norm set by authorities in the field (e.g. SART). For example, SART researchers also do not share data in their research publications.

Importantly, Univfy and the co-authors do not own the data so we do not have the authority to make decisions about data-sharing.

Speaking only out of an educated guess and an acute awareness of the sensitivity around reproductive health, the reasons are likely complex. Patients may perceive risks from having sought care at a fertility center which shares data publicly. This sensitivity, perceived or possibly real risk may be very different in other disease areas such as oncology. Patients may not feel comfortable seeking care at a fertility center that shares data publicly even if de-identified.

We hope that reviewers will take into consideration the following attributes of Univfy and related challenges:

- 1) Univfy is a small, social mission-driven, for-profit company that depends on fundraises. Univfy was founded based on academic research funded by a non-profit (Coulter Foundation) which believed in disseminating technology in the marketplace to expedite its reach to providers and patients.
- 2) The issues raised by this reviewer regarding proprietary framework and product are based on licensing with the original academic center and affect the entire health/life sciences ecosystem.
- 3) We have made our best effort to collaborate and support others' research using our platform as evidenced by past publications. We also make our best effort to share back insights in the form of peer-reviewed research as much as possible. Example: Yao et al., Semin Reprod Med 2024. Patient-centric in vitro fertilization prognostic counseling using

machine learning for the pragmatist. <https://www.thieme-connect.com/products/ejournals/html/10.1055/s-0044-1791536>

- 4) In the past, public funding was very limited at best in supporting this type of research. We conduct private fundraising to support ongoing R&D.